

Teams Rooms Project Instruction Manual

Oct 2, 2026 · Version 1.0

Every wave in the Teams Rooms project runs the same way: prepare, check, change one account, test, observe and record. Gate criteria, acceptance tests and recovery tables are defined in [Doc](#) ; the instructions here show how to carry them out.

How to use this manual

Find the activity in hand below, follow its instructions, and check the pass criteria in the runbook section it names.

Document	Use it for
CAB submission	Scope, approved decisions, wave sizes, risks and backout limits
Passwordless runbook	Gate 0 checklist, supported policies, per-account conversion, acceptance tests, device recovery
Cloud identity runbook	Gate 1 checklist, baseline register, RoomList and account transfer, rollback, AD retirement
This manual	Setting up, running windows, recording evidence, communicating and escalating
Change record	Named owners, window dates, phase releases, exceptions and the CAB decision

Role	Responsibilities in a window
Change owner	Go or no-go, phase release, stop and resume decisions, communications
Room engineer	Pre-checks, device work, acceptance tests, evidence capture
Teams Administrator	Pro Management migration, remote sign-out, update pauses
Identity engineer	Password cleanup, authority transfer, sync and audit checks

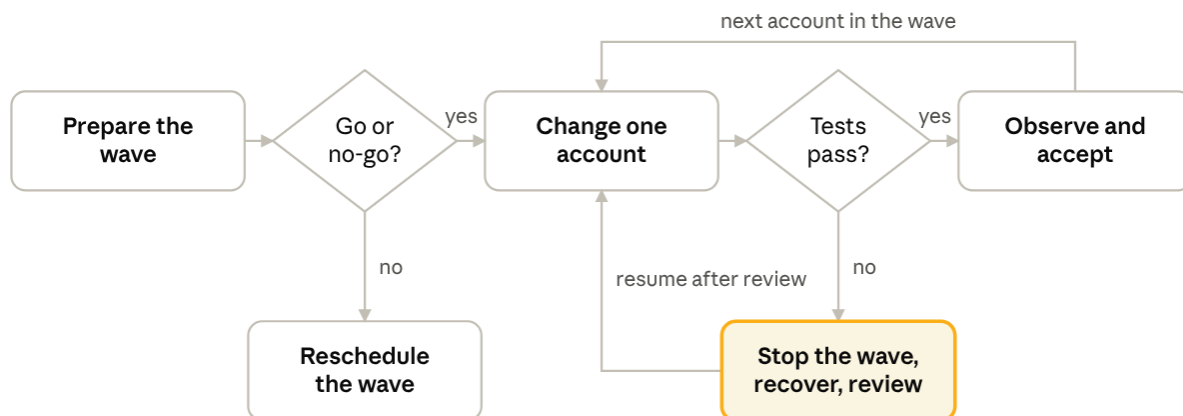
Role	Responsibilities in a window
Exchange engineer	Mailbox and RoomList baselines, RoomList membership changes
Security owner	Policy exceptions, supervised rollback windows
Site contact	Physical access, restarts, sign-in at the device

One person can hold several roles. Where staffing allows, the change owner and the person executing the change are different people.

The change-window loop

Every window, passwordless or identity, follows the same loop: one account at a time, tested before the next begins.

Every window repeats one loop, and a failed test stops the wave



change-window loop · 5 steps, 2 decisions

A failed test sends the wave to stop, recover and review; only the change owner's recorded decision returns it to the loop.

One-time setup

Complete these steps once, before the Gate 0 review. Each produces evidence for the readiness pack.

1. **Access.** Make the project roles eligible in Privileged Identity Management, activated per window: Teams Administrator, User Administrator, Exchange Administrator and Hybrid Identity Administrator. Global Administrator is eligible only for named people, for supervised rollback.

2. **Administrative unit.** In the Entra admin centre, create an administrative unit holding only the approved room accounts and project groups. Assign Hybrid Identity Administrator scoped to that unit.
3. **Graph consent.** An Application Administrator grants the chosen tool (Microsoft Graph Command Line Tools or the project app registration) these delegated scopes: User-OnPremisesSyncBehavior.ReadWrite.All, Group-OnPremisesSyncBehavior.ReadWrite.All, User.Read.All, GroupMember.Read.All, AuditLog.Read.All and OnPremDirectorySynchronization.Read.All.
4. **Tools.** Install the modules on the admin workstation. The ActiveDirectory module is used from a management server with RSAT.

```
Install-Module ExchangeOnlineManagement -Scope CurrentUser
Install-Module Microsoft.Graph.Authentication, Microsoft.Graph.Users,
Microsoft.Graph.Groups, Microsoft.Graph.Identity.DirectoryManagement,
Microsoft.Graph.Reports -Scope CurrentUser
```

5. **Evidence location.** Create the restricted evidence folder, structured as described under Evidence and records, with access limited to project roles.
6. **Tenant baseline.** Record the directory-sync configuration ID and the hard-match protection value. Rollback depends on both.

```
Connect-MgGraph -Scopes 'OnPremDirectorySynchronization.Read.All'
Get-MgDirectoryOnPremiseSynchronization |
    Select-Object Id, @{n='HardMatchBlocked'; e=
    {$_.Features.BlockCloudObjectTakeoverThroughHardMatchEnabled}}
```

7. **Sync servers.** On the active and staging Connect Sync servers, record the installed version from Programs and Features, then record the scheduler state.

```
Get-ADSyncScheduler | Select-Object SyncCycleEnabled, StagingModeEnabled,
CurrentlyEffectiveSyncCycleInterval
```

8. **Recovery kit.** Reserve a spare Android device and a spare or scheduled Windows room for rehearsals. Stage the OEM recovery image or Microsoft recovery tool media, and record local administrator access for each Windows room.

Manifest and baselines

The inventory script captures mailbox, calendar processing, permission and Places data for the approved list. The commands below add the identity, AD, RoomList and Windows fields the runbooks also need.

Work in this order: approved CSV, inventory script, the commands below, manual fields, then one reconciled register row per account. A failed query is entered as *Unknown*, never left blank.

Identity and sync state (Graph PowerShell)

```
Connect-MgGraph -Scopes 'User.Read.All','User-
OnPremisesSyncBehavior.ReadWrite.All'
$rooms = Import-Csv .\approved-rooms.csv
$identity = foreach ($r in $rooms) {
    $u = Get-MgUser -UserId $r.UPN -Property
    Id,UserPrincipalName,OnPremisesSyncEnabled,OnPremisesImmutableId,OnPremisesSe
    curityIdentifier,LicenseAssignmentStates
    $soa = Invoke-MgGraphRequest -Method GET -Uri
    "https://graph.microsoft.com/v1.0/users/$(($u.Id)/onPremisesSyncBehavior?
    '$select=isCloudManaged"
    [pscustomobject]@{
        RoomId          = $r.RoomId
        UPN              = $u.UserPrincipalName
        ObjectID         = $u.Id
        SyncEnabled      = $u.OnPremisesSyncEnabled
        ImmutableId      = $u.OnPremisesImmutableId
        Sid              = $u.OnPremisesSecurityIdentifier
        IsCloudManaged  = $soa.isCloudManaged
        LicenceGroups    = ($u.LicenseAssignmentStates | Where-Object
        AssignedByGroup).AssignedByGroup -join ';'
    }
}
$identity | Export-Csv .\evidence\baseline\identity.csv -NoTypeInfo
```

AD anchors and Exchange attributes (ActiveDirectory module)

```
$ad = foreach ($r in $rooms) {
    Get-ADUser -Filter "UserPrincipalName -eq '$($r.UPN)'" -Properties
    ObjectGUID,ObjectSid,'mS-DS-
    ConsistencyGuid',msExchRemoteRecipientType,msExchRecipientTypeDetails,targetA
    ddress,proxyAddresses |
    Select-Object UserPrincipalName, ObjectGUID, ObjectSid,
    DistinguishedName,
        @{n='ConsistencyGuid'; e={ if ($_.mS-DS-ConsistencyGuid') {
    [guid]::new([byte[]]$_.'mS-DS-ConsistencyGuid') } }},
        msExchRemoteRecipientType, msExchRecipientTypeDetails, targetAddress,
        @{n='ProxyAddresses'; e={ $_.proxyAddresses -join ';' }}
}
$ad | Export-Csv .\evidence\baseline\ad-anchors.csv -NoTypeInfo
```

RoomLists (Exchange Online PowerShell)

```
Connect-ExchangeOnline
Get-DistributionGroup -RecipientTypeDetails RoomList -ResultSize Unlimited |
  Select-Object Name, PrimarySmtpAddress, ExternalDirectoryObjectId,
  IsDirSynced, HiddenFromAddressListsEnabled,
  @{n='ManagedBy'; e={ $_.ManagedBy -join ';' }},
  @{n='Members'; e={ (Get-DistributionGroupMember $_.Identity -ResultSize
  Unlimited).PrimarySmtpAddress -join ';' }} |
  Export-Csv .\evidence\baseline\roomlists.csv -NoTypeInfo
```

Windows room PC (run in admin mode on each of the five PCs)

```
$cv = Get-ItemProperty 'HKLM:\SOFTWARE\Microsoft\Windows NT\CurrentVersion'
"Edition $($cv.EditionID); version $($cv.DisplayVersion); build
$($cv.CurrentBuild).$($cv.UBR)"
Get-Tpm | Select-Object TpmPresent, TpmReady
dsregcmd /status | Select-String 'AzureAdJoined|DomainJoined|TenantName'
netsh winhttp show proxy
```

Manual fields. Record the OEM and model, Teams Rooms app version and Pro Management eligibility from the Migration tab. For Android devices, record model, serial, MAC and live app versions from the Pro Management device page.

Running a passwordless wave

A production wave takes about a week from first notice to cleanup. The account-level steps are the passwordless runbook's per-account procedure.

When	Action	Owner
5 working days before	Confirm the wave's accounts and endpoints in the register; re-run the baseline; send the wave notice; reserve rooms and panels	Change owner
2 working days before	Recheck Pro Management eligibility and live versions; pause any Android version due during the window; confirm the site contact	Room engineer, Teams Administrator
1 working day before	Go or no-go review of the Gate 0 items for these accounts; record each room's backout decision point; request role activation	Change owner

When	Action	Owner
Window start	Activate roles; confirm no meeting is in progress; open the wave evidence record	Room engineer
During the window	Convert one account at a time, following runbook steps 1 to 8	Teams Administrator, room engineer
Window end	Publish availability, or declare an overrun	Change owner
Next business day	Review sign-ins, compliance and incidents per account; accept or hold the wave	Change owner
Cleanup day	Scramble each accepted account's password; restart and retest	Identity engineer, room engineer
After cleanup	Resume paused Android versions; record completion; brief the next wave	Teams Administrator, change owner

The pilot differs in two ways: it holds 24 hours before cleanup, and five accepted business days after cleanup before any production wave starts.

Pausing an Android version. In Pro Management, open **Updates > Update management > Android updates**, select the version and pause it. The tenant-wide **Pause updates** action stops all Android updates for two weeks and needs change-owner approval.

Windows waves. On-site support is booked for the whole window, with recovery media and local administrator access to hand.

Next-day sign-in review. This returns interactive sign-ins; non-interactive sign-ins are reviewed in the Entra admin centre sign-in logs.

```
Connect-MgGraph -Scopes 'AuditLog.Read.All'
$since = (Get-Date).ToUniversalTime().AddDays(-1).ToString('yyyy-MM-ddTHH:mm:ssZ')
Get-MgAuditLogSignIn -All -Filter "userPrincipalName eq 'room01@example.com'
and createdDateTime ge $since" |
    Select-Object CreatedDateTime, AppDisplayName, ConditionalAccessStatus,
        @{n='ErrorCode'; e={$_.Status.ErrorCode }}, @{n='Reason'; e={$_.Status.FailureReason }}
```

Running a cloud identity wave

Each identity wave converts one RoomList, then its member accounts one at a time. The account steps are the cloud identity runbook's account transfer and validation procedure.

When	Action	Owner
5 working days before	Select the RoomList and its member accounts; confirm no members outside scope; re-run baselines; send the wave notice	Change owner
1 working day before	Confirm Gate 1 still holds, no AD writer has touched the accounts, and the direct licence fallback is ready	Identity engineer
Window, first part	Convert the RoomList; run a sync; verify subtype, membership and Room Finder	Identity engineer, Exchange engineer
Window, second part	Transfer member accounts one at a time and verify each	Identity engineer
Window end	Restart and test the rooms; publish availability	Room engineer
Next business day	Review sign-ins; accept or hold the wave	Change owner

Convert the RoomList, then each account (Graph PowerShell, scoped Hybrid Identity Administrator)

```
Connect-MgGraph -Scopes 'Group-OnPremisesSyncBehavior.ReadWrite.All','User-OnPremisesSyncBehavior.ReadWrite.All'
Invoke-MgGraphRequest -Method PATCH -Uri
"https://graph.microsoft.com/v1.0/groups/$groupId/onPremisesSyncBehavior" -
Body @{ isCloudManaged = $true }
# after the RoomList checks pass, per account:
Invoke-MgGraphRequest -Method PATCH -Uri
"https://graph.microsoft.com/v1.0/users/$userId/onPremisesSyncBehavior" -Body
@{ isCloudManaged = $true }
```

Run a delta sync (active Connect Sync server)

```
Start-ADSyncSyncCycle -PolicyType Delta
```

Check the RoomList (Exchange Online). Expect RecipientTypeDetails of RoomList, IsDirSynced of False, and no output from the comparison.

```
Get-DistributionGroup -Identity $roomList | Select-Object
RecipientTypeDetails, IsDirSynced, PrimarySmtpAddress
$was = ((Import-Csv .\evidence\baseline\roomlists.csv | Where-Object
PrimarySmtpAddress -eq $roomList).Members) -split ';'
$now = (Get-DistributionGroupMember $roomList -ResultSize
Unlimited).PrimarySmtpAddress
Compare-Object $was $now
```

Check each account. Expect isCloudManaged of True, an empty onPremisesSyncEnabled, and one audit entry. Reading audit logs also needs a directory role that can read them, such as Reports Reader.

```
(Invoke-MgGraphRequest -Method GET -Uri
"https://graph.microsoft.com/v1.0/users/$userId/onPremisesSyncBehavior?
`$select=isCloudManaged").isCloudManaged
(Get-MgUser -UserId $userId -Property
OnPremisesSyncEnabled).OnPremisesSyncEnabled
Get-MgAuditLogDirectoryAudit -Filter "activityDisplayName eq 'Change Source
of Authority from AD to cloud' and targetResources/any(t:t/id eq '$userId')"
|
    Select-Object ActivityDateTime, Result, @{n='By'; e={
$_ .InitiatedBy.User.UserPrincipalName }}
```

On the sync server, the object's Microsoft Entra connector entry shows blockOnPremisesSync as true (Synchronization Service Manager, Connectors, search, Lineage). Event 6956 is logged only when a later AD change is blocked, so its absence is not a failure.

Room Finder is checked by hand in Outlook and Teams: the list appears with every member room, and a direct booking succeeds.

Evidence and records

Evidence is filed by phase, wave and account, so every room can be traced from baseline to retirement.

```

TeamsRooms-Project/
  00-Readiness/      manifest, Gate 0 pack, tenant baseline, rehearsal
results
  01-Baseline/       identity.csv, ad-anchors.csv, roomlists.csv, inventory
output
  02-Passwordless/
    Pilot/<RoomId>/
    Wave-01/<RoomId>/ ... to Wave-08
  03-Groups/         RoomList before and after exports, audit references
  04-Identity/
    Pilot/<RoomId>/
    Wave-NN/<RoomId>/
  05-Retirement/    dependency checks, retirement actions, post-retirement
tests
  06-Closure/        handover sign-off, incident comparison

```

Each account folder holds the before and after exports, the Pro Management job result, test results, the sign-in export and any audit reference. Files are named

`<RoomId>_<phase>_<item>_<yyyymmdd>` , for example `LOCAL-001_PWL_tests_20261020.csv` .

Register columns

Column	Content
RoomId, UPN, ObjectId	From the approved manifest
Platform, site	Android or Windows; local or remote
Endpoints	Count, models and device IDs on the account
Waves	Passwordless wave and identity wave
Passwordless dates	Converted, accepted, password cleaned
Cloud authority accepted	Date and audit reference
AD independence accepted	Date and retirement action
Exceptions	Change-record reference, owner and expiry

Passwords, tokens and verification codes are never recorded. Screenshots are cropped before filing.

When something goes wrong

A stop condition pauses the rest of the wave at once, and only the change owner resumes it.

1. **Stop.** Leave the affected account in its current state, pause the remaining accounts and note the time.
2. **Preserve.** Export the Pro Management job detail, sign-in logs and device state before changing anything.
3. **Inform.** Give the change owner the room, account, symptom and time.
4. **Decide.** The change owner compares the clock with the backout decision point. A fix forward is taken only if the cause is understood and fits; otherwise the runbook recovery is used.
5. **Recover.** Follow the platform recovery in the passwordless runbook, or the rollback in the cloud identity runbook.
6. **Escalate.** If recovery may pass the window end, raise an incident, ask facilities to tell the room's bookers and offer an alternative room.
7. **Review.** Before resuming, the change owner reviews the evidence and records the decision in the change record.

Situation	Escalate to
Device will not sign in after recovery	OEM support and a Microsoft support case
Unexpected deletion, duplicate object or lost membership	Identity and security owners at once; all identity waves frozen
Conditional Access or compliance block	Security owner
Booking or RoomList fault	Exchange owner
Recovery passing the window end	Service desk incident and facilities

Supervised rollback window. Used only for identity-authority rollback. The change owner, identity engineer, security owner and Global Administrator are on the same call, and no other sync-scope change runs in the tenant meanwhile.

1. The security owner opens the window and records the start time.
2. The Global Administrator relaxes hard-match protection.
3. The identity engineer completes the account rollback steps in the cloud identity runbook.
4. The Global Administrator restores the value recorded at setup and confirms it, whether or not the takeover succeeded.

5. The security owner records the end time and final value, and closes the window.

```
Connect-MgGraph -Scopes 'OnPremDirectorySynchronization.ReadWrite.All'
$s = Get-MgDirectoryOnPremiseSynchronization
# step 2: relax
$s.Features.BlockCloudObjectTakeoverThroughHardMatchEnabled = $false
Update-MgDirectoryOnPremiseSynchronization -
OnPremisesDirectorySynchronizationId $s.Id -Features $s.Features
# step 4: restore the recorded value, then confirm
$s.Features.BlockCloudObjectTakeoverThroughHardMatchEnabled = $true
Update-MgDirectoryOnPremiseSynchronization -
OnPremisesDirectorySynchronizationId $s.Id -Features $s.Features
(Get-
MgDirectoryOnPremiseSynchronization).Features.BlockCloudObjectTakeoverThrough
HardMatchEnabled
```

Communications templates

Four messages cover every wave; replace each bracketed field before sending.

Wave notice, to room users and organisers, five working days before

Subject: Meeting room maintenance on [date]: [room names]

The rooms below are reserved for maintenance on [date] from [start] to [end]:
[room list]

The rooms cannot be used during this time. Meetings booked in that window
have been moved to [alternative rooms].

After the work, each room's Teams account leaves the meeting chat when a
meeting ends and no longer has access to files or recordings shared in that
meeting. Your own access is unchanged.

Questions: [service desk contact], quoting [change reference].

Service desk briefing, internal, two working days before

Subject: Teams Rooms change [change reference]: [date]

Rooms: [room list]

Window: [date], [start] to [end]

Expected after the change: rooms sign in automatically after restarts; room accounts leave meeting chats after meetings.

Route to the room engineer on call ([contact]) if a room shows a sign-in screen, panels show wrong availability, or a booking fails.

Do not sign out, reset or re-enrol a room device.

Back in service, to the same audience as the notice

Subject: Meeting rooms back in service: [room names]

Maintenance on [room names] finished at [time]. All rooms passed testing and are available to book.

If anything does not work as expected, contact [service desk contact] quoting [change reference].

Overrun, sent as soon as the change owner declares it

Subject: Meeting room maintenance extended: [room name]

Work on [room name] is taking longer than planned. The room stays unavailable until [time].

Please use [alternative room] until then. Facilities will contact organisers with affected bookings directly.

Updates: [service desk contact], quoting [change reference].

Closure and handover

The project closes when every item below is ticked and the change owner records closure in the change record.

- ☐ Every register row shows passwordless, cloud authority and AD independence accepted, or a CAB-approved exception.
- ☐ Hard-match protection confirmed at the value recorded during setup.
- ☐ Listed AD objects retired, and every room retested afterwards.
- ☐ Paused Android versions resumed; no project pause left in Pro Management.

- ☐ Project role eligibility removed, administrative unit scope reviewed, and Graph consents removed where no longer needed.
- ☐ Operations accept the cloud password reset and device recovery procedure, including replacement devices: password set-up first, then conversion.
- ☐ Operations accept RoomList membership changes in Exchange Online, plus licence and Conditional Access monitoring for room accounts.
- ☐ Spare devices and recovery media returned to stock, with their state recorded.
- ☐ Authentication incidents and site visits compared with the pre-change baseline, and the result filed in 06-Closure.
- ☐ Evidence folder access reduced to the retention owner under the organisation's retention policy.

Quick reference

The locations, calls and numbers used most often during the project.

Task	Where or how
Convert an account	Pro Management: Planning > Resource Accounts > Migration > Schedule migration
Scramble a cloud-managed password	Pro Management: Planning > Resource Accounts > Migration > Cleanup password
Apply Set as Resource on its own	Pro Management: Planning > Resource Accounts > select accounts > Set as Resource
Pause one Android version	Pro Management: Updates > Update management > Android updates > select the version
Read authority state	GET /v1.0/users/{id}/onPremisesSyncBehavior? \$select=isCloudManaged
Transfer authority	PATCH /v1.0/users/{id}/onPremisesSyncBehavior with isCloudManaged: true (groups use /groups/{id}/)
Audit activities	<i>Change Source of Authority from AD to cloud; Undo changes to Source of Authority from AD DS to cloud</i>
Run a delta sync	Start-ADSyncSyncCycle -PolicyType Delta

Task	Where or how
Windows join state	<code>dsregcmd /status</code> on the room PC

Item	Value
Estate	50 rooms: 35 local Android, 10 remote Android, 5 local Windows
Pilot	4 rooms: 2 local Android, 1 remote Android, 1 local Windows
Production waves	3 of 11 local Android, 3 of 3 remote Android, 2 of 2 Windows
Passwordless pilot hold	24 hours before cleanup, 5 accepted business days after
Production wave acceptance	1 business day of normal use
Identity pilot hold	5 accepted business days
Retention before AD retirement	30 calendar days after the final identity wave
Android auto-update window	00:00 to 05:00 device-local time
Tenant-wide Android pause	2 weeks
Connect Sync deadline	2.6.84.0 or later with application-based authentication by 7 April 2027